THE QUANTUM-CLASSICAL PORTFOLIO

A HYBRID FRAMEWORK FOR EXPLAINABLE OPTIMISATION UNDER REAL-WORLD FINANCIAL CONSTRAINTS

Bridging the gap between combinatorial quantum mathematics and continuous classical asset allocation.

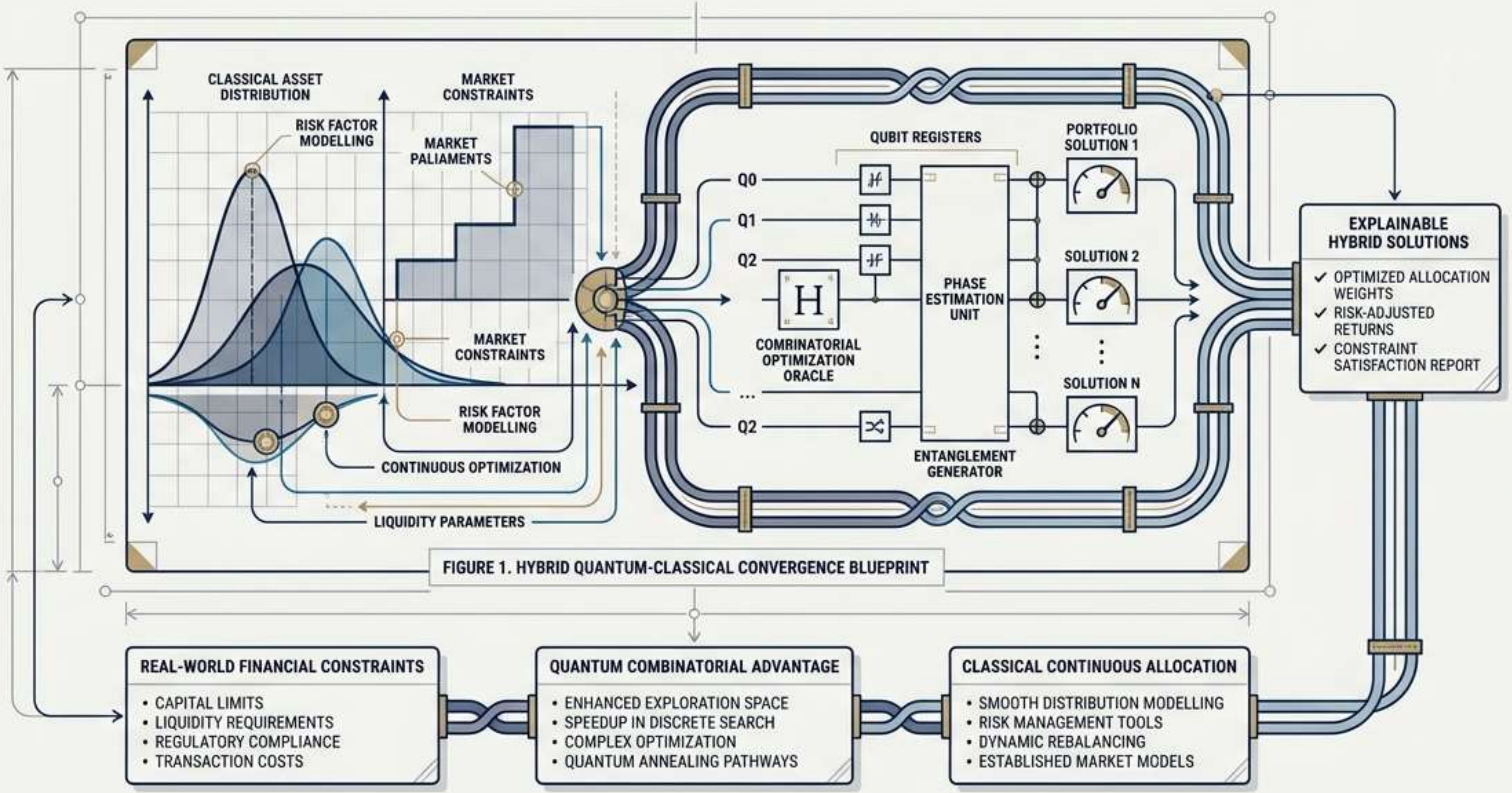


FIGURE 1. HYBRID QUANTUM-CLASSICAL CONVERGENCE BLUEPRINT

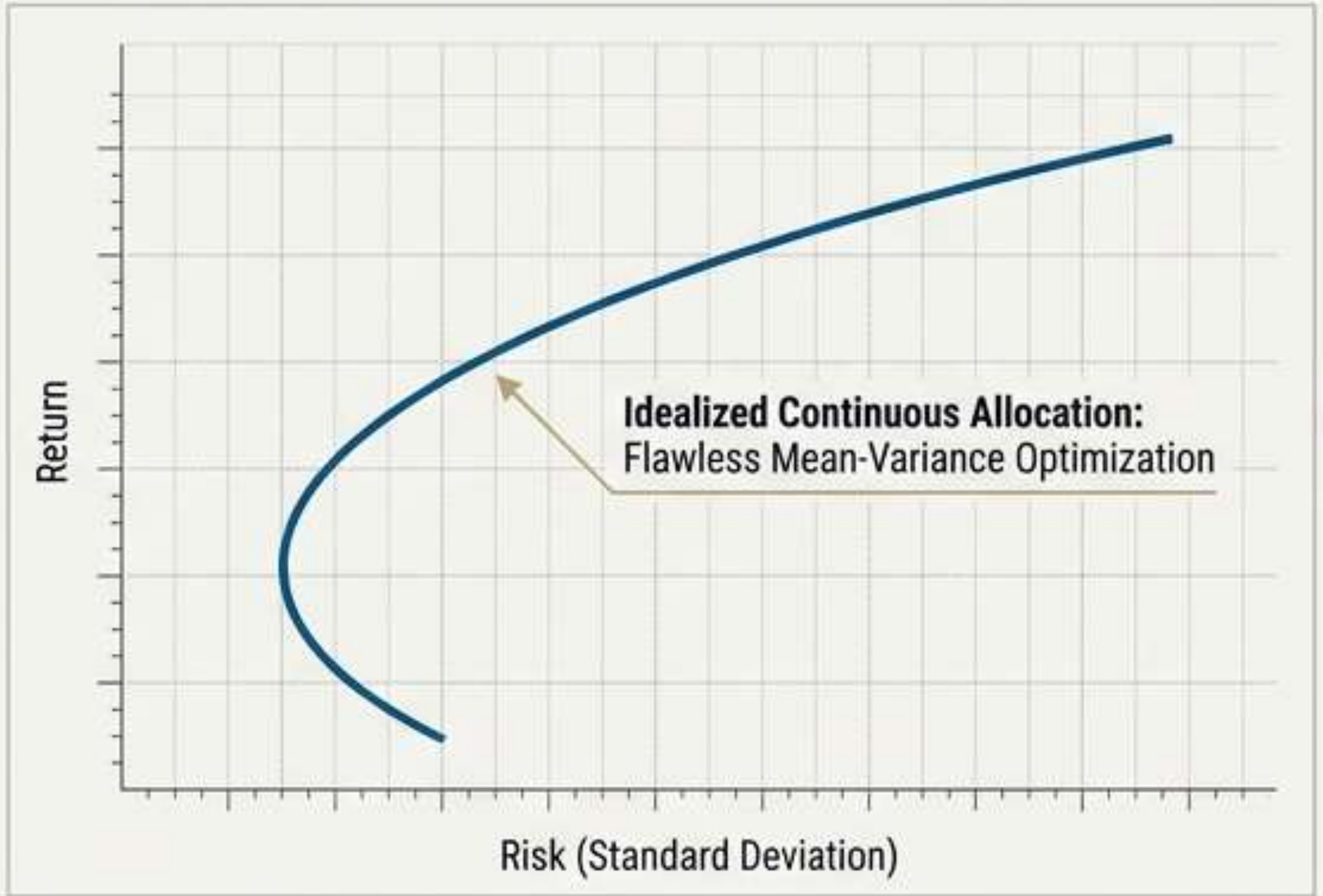
NotebookLM

The Bottleneck of Reality

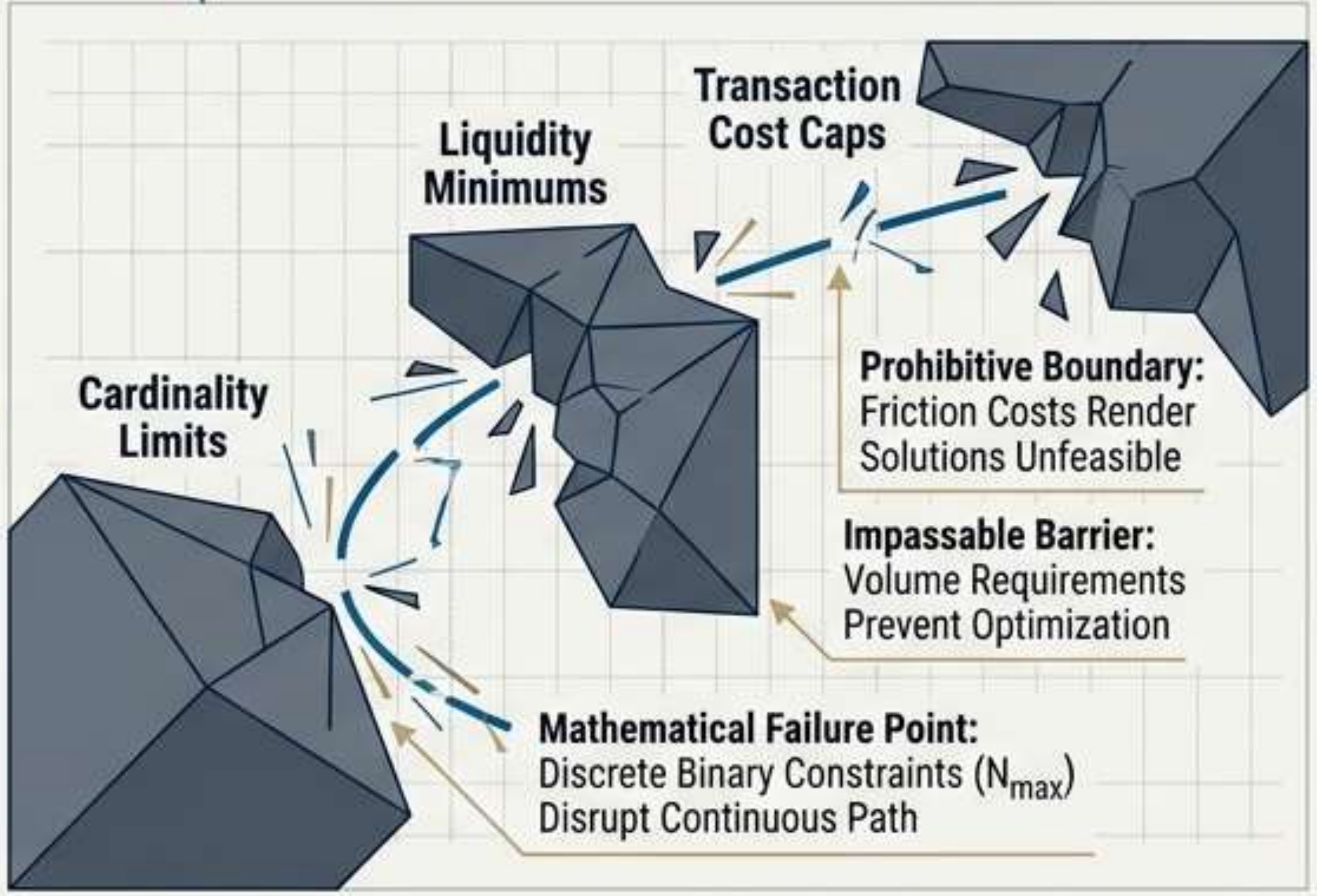
The Markowitz mean-variance framework works flawlessly for continuous allocations.
However, adding discrete, binary constraints transforms portfolio construction into a Mixed-Integer Quadratic Program (MIQP).

The Scaling Wall: As candidate asset universes grow, this combinatorial search space becomes computationally prohibitive for classical computers.

Classical Theory



Real-World Constraints



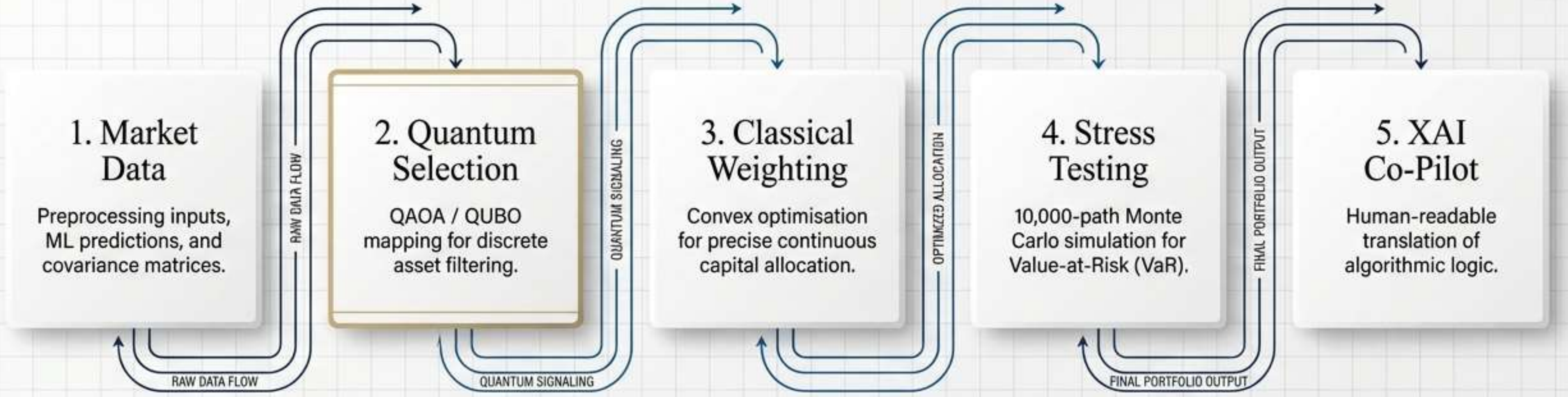
The Optimisation Paradigm Matrix

Finding the pragmatic path between classical scale and quantum capabilities.

	Pure Classical	Pure Quantum	Hybrid Approach
Handling Discrete Constraints	✗ Fails to scale (MIQP)	✓ Native handling via QAOA	✓ Quantum-assisted selection
Handling Continuous Allocation	✓ Highly efficient	✗ Inefficient	✓ Classical Convex Optimisation
Hardware Readiness	✓ Ready	⚠ Limited by NISQ errors	✓ Deployable today
Explainability	✓ High	✗ Low (Black box)	✓ AI Portfolio Co-Pilot enabled

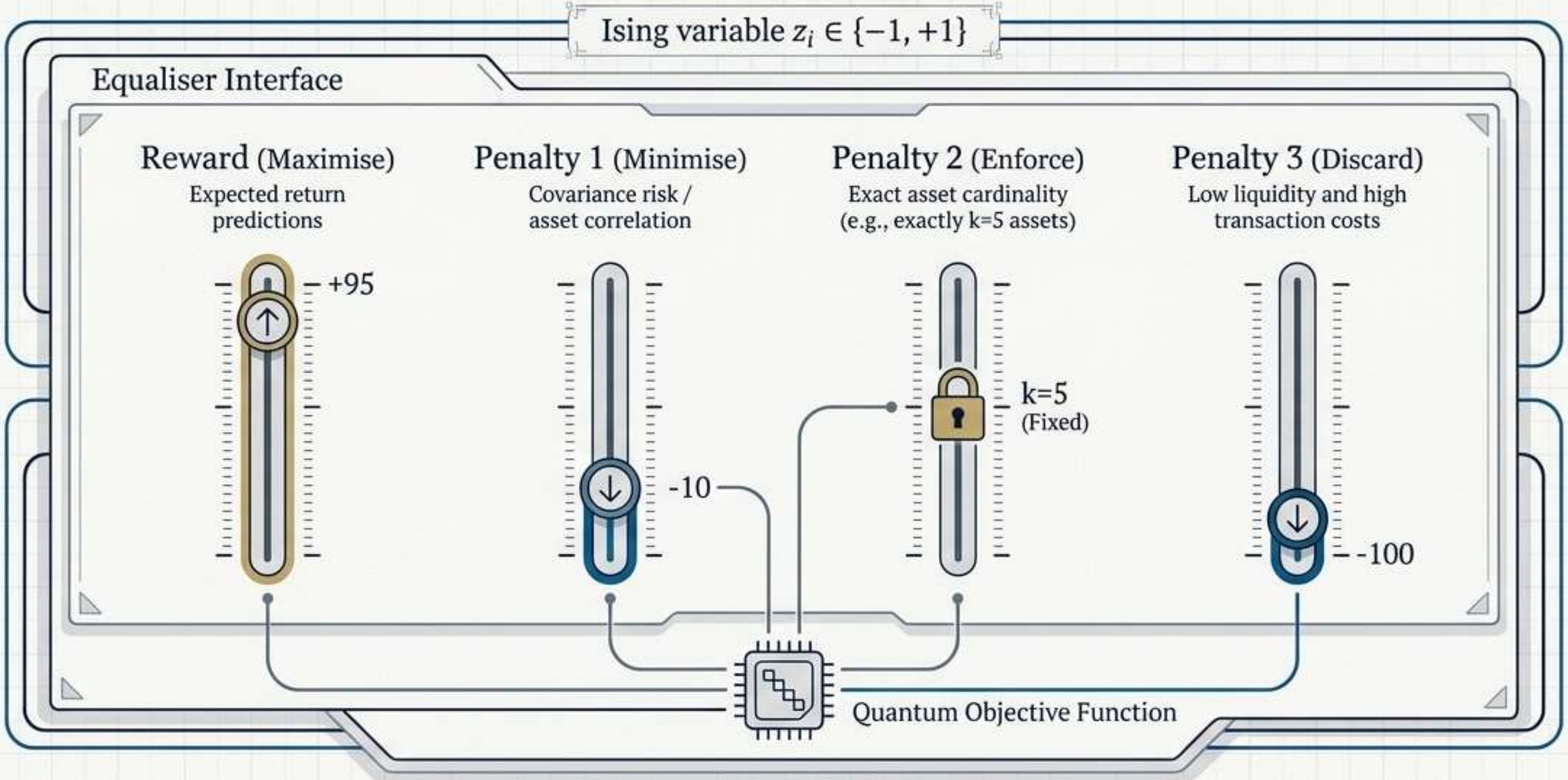
The 5-Stage Blueprint

A hybrid system architecture refining chaotic market variables into a structured, transparent portfolio.



Formulating the Quantum Objective

To harness quantum compute, financial goals must be mapped into an Ising Hamiltonian. The QUBO matrix encodes four competing objectives:



The Hybrid Hand-off: Discrete to Continuous

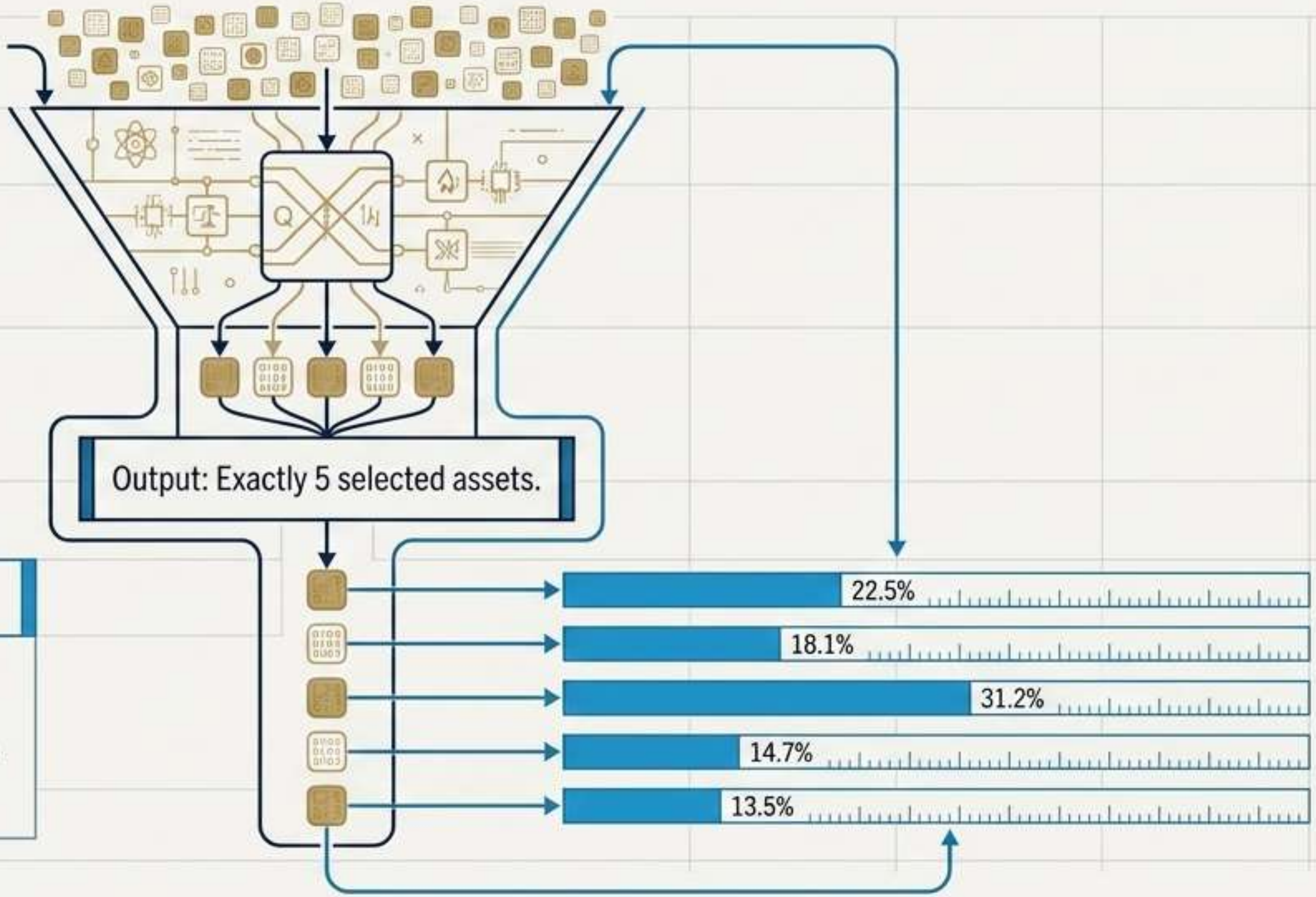
The Synergy: Quantum mathematics does the heavy lifting of selection; classical mathematics ensures the precision of allocation.

The Quantum Realm

QAOA evaluates massive combinatorial spaces of discrete constraints.
Outputs binary decisions (x_i = 1 or 0).

The Classical Realm

Selected subset enters a classical convex optimiser. Solves for continuous variables (w_i), determining exact capital allocation percentages while preventing short-selling.



Beyond the Point Estimate: Monte Carlo Stress Testing

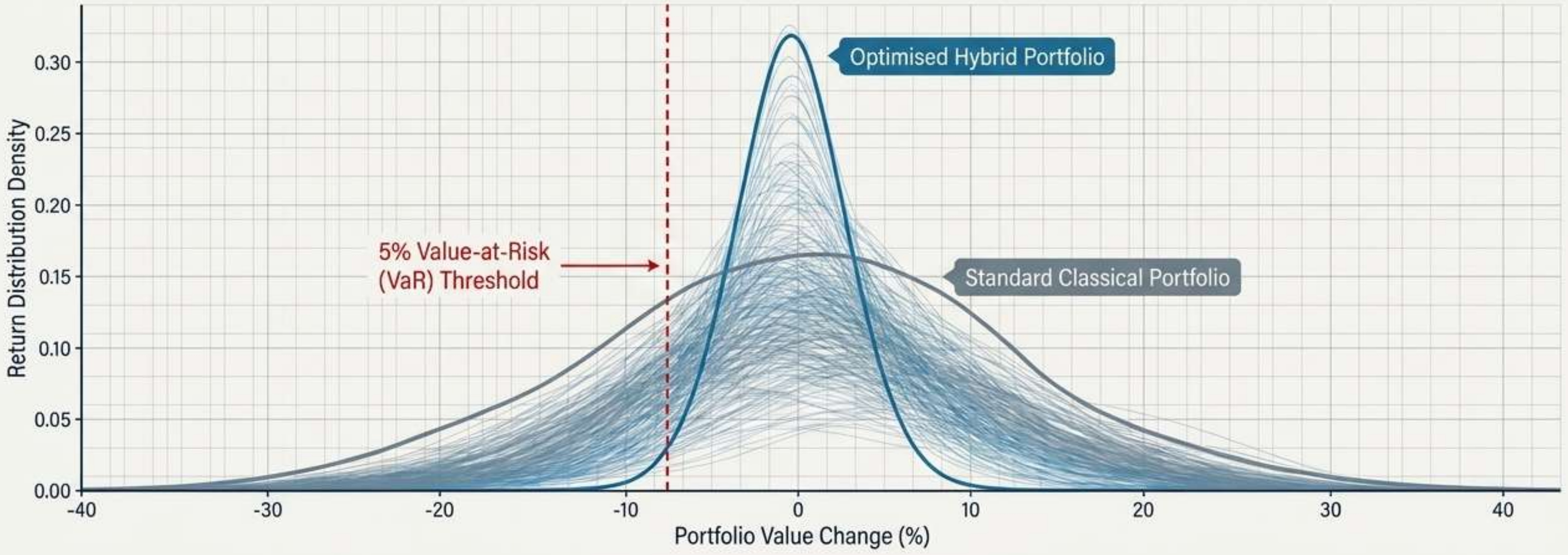
Mean-variance analysis alone is insufficient for assessing downside robustness.

The Methodology:

The framework executes 10,000 independent simulation paths to evaluate empirical downside risk under adverse conditions.

The Metric:

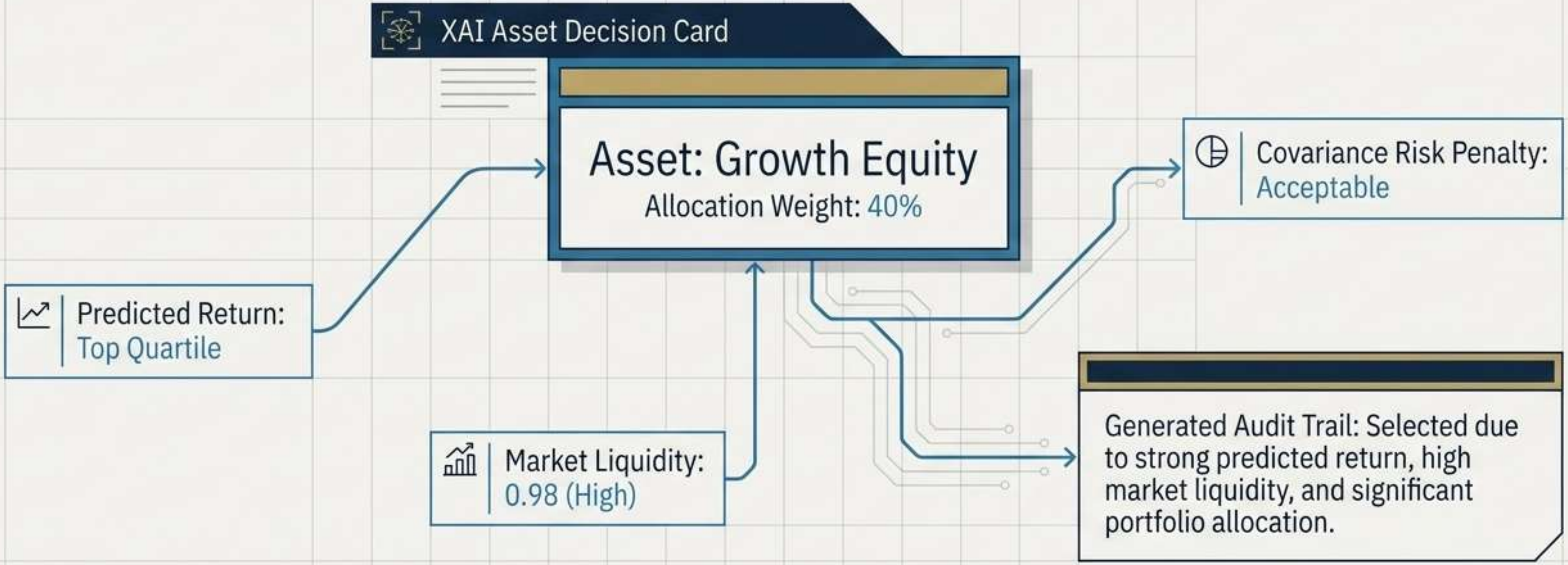
5% Value-at-Risk (VaR)—estimating maximum expected loss over the investment horizon with 95% confidence.



The AI Portfolio Co-Pilot

Algorithmic optimisation is useless if regulatory stakeholders and portfolio managers cannot audit it.

Rule-Based Translation: The XAI layer maps optimisation outcomes against primary criteria (return, liquidity, diversification).



The Financial Scorecard: Maximum Efficiency

The Trade-Off: The hybrid framework willingly trades a fractional drop in theoretical maximum return for a substantial plunge in overall risk exposure.

	Classical Benchmark	Hybrid Quantum-Classical	
Expected Return	0.0016	0.0015 (Slight nominal drop)	
Portfolio Risk (Volatility)	0.0127	0.0084	34% Decrease
Sharpe Ratio	0.1284	0.1765	Massive Efficiency Gain

Operational Reality: **Cheaper and Easier to Trade**

The Insight: By encoding real-world constraints directly into the QUBO objective, the quantum-assisted portfolio structurally favours highly liquid assets with low trading costs, yielding a more actionable deployment.

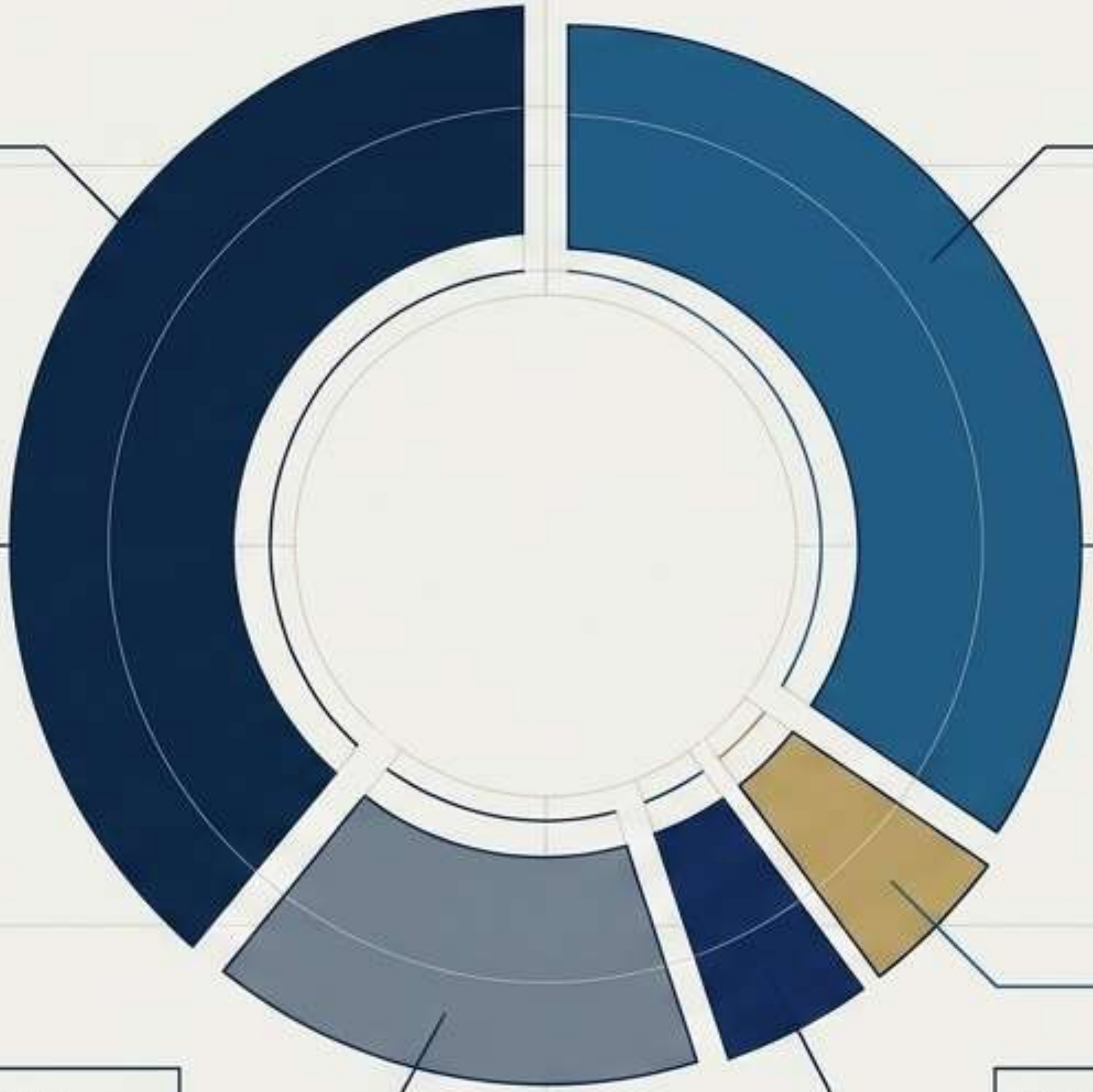
Zero Constraint Violations

(QAOA Objective: -49.3765)

Anatomy of the Optimised Portfolio

Growth Equity (40%): The anchor asset (highest predicted return, acceptable costs).

International Bonds (34.25%): The stabiliser (low correlation to equities).



Value Equity (15.75%): Return enhancer with high baseline liquidity.

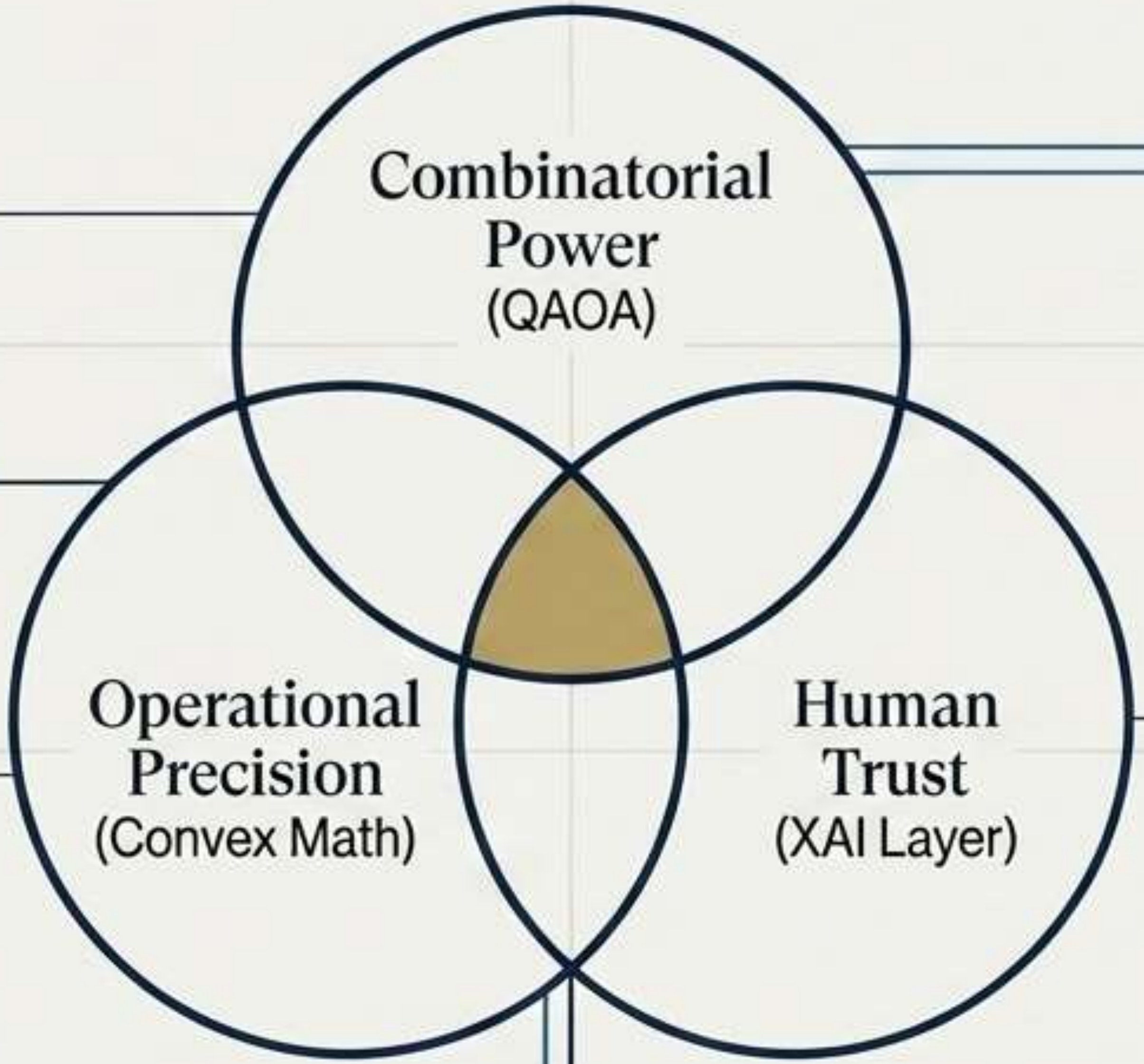
Govt Bonds (5%) & Domestic Equity (5%): Pure diversification plays enforced to satisfy rigorous covariance risk penalties.

The Risk-Return-Transparency Nexus

The triumph of this framework is not quantum supremacy. It is practical synergy.

By shifting the discrete selection burden to QAOA, maintaining continuous weighting in classical compute, and wrapping the output in natural language, we achieve structurally sound portfolios that humans can actually understand and deploy today.

Discrete Selection Burden:
Shifted to QAOA.
Optimized for large-scale combinatorial problem space.



Continuous Weighting:
Maintained in Classical Compute. Ensures efficient and precise allocation within defined constraints.

Natural Language Wrapping:
Output interpreted for human understanding.
Provides explainable and actionable portfolio insights.

Current Limitations of the Framework

Hardware Bottlenecks

NISQ devices restrict qubit counts and circuit depth, limiting the absolute size of the candidate asset universe.

Parameter Sensitivity

QUBO penalty hyperparameters (α, β, γ, δ) require meticulous prior calibration to avoid suboptimal combinations.

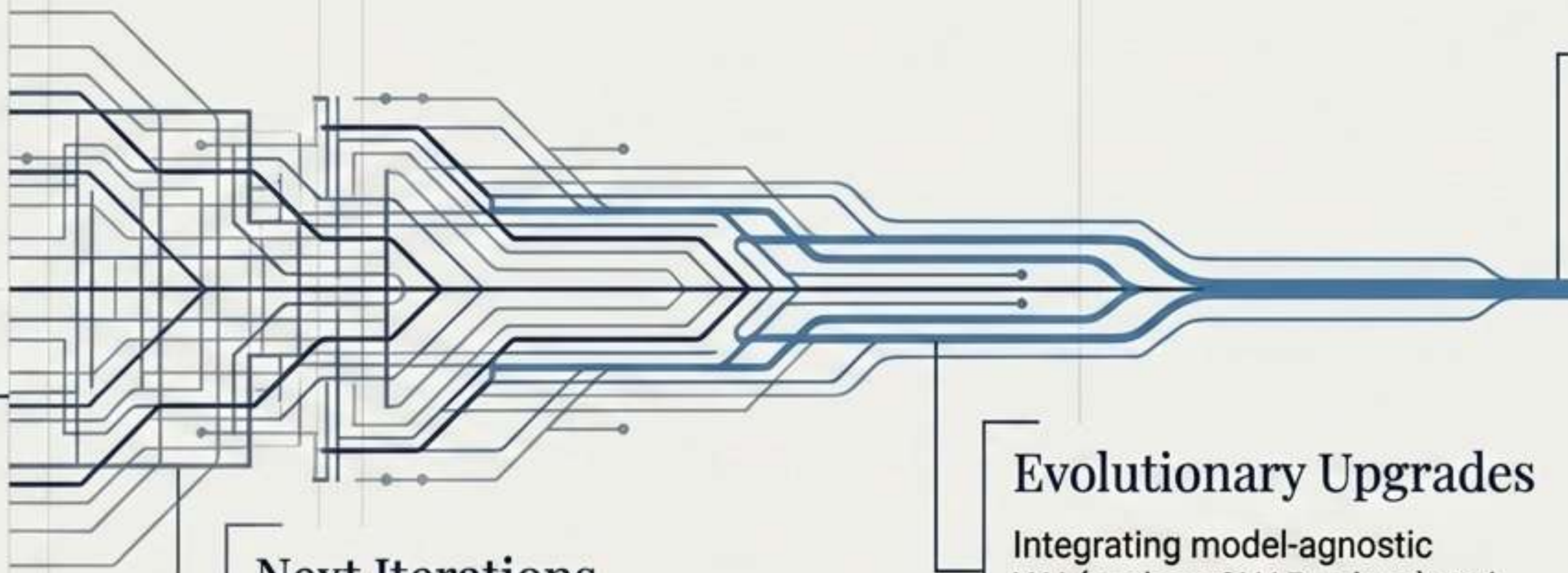
Market Assumptions

The Monte Carlo module assumes normal distributions, potentially underrepresenting heavy-tail market events.

XAI Maturity

Current explanations are rule-based and deterministic rather than utilising advanced attribution models like SHAP.

The Future of Quantum Finance



The Horizon

This framework establishes the foundational pipeline for institutional adoption the moment fault-tolerant quantum hardware scales.

Evolutionary Upgrades

Integrating model-agnostic XAI (such as SHAP values) and multi-period portfolio rebalancing capabilities.

Next Iterations

Expanding dataset processing, rigorous out-of-sample backtesting, and deployment on advanced, lower-noise quantum hardware.

Conclusion: A Pragmatic Pathway

The Hybrid Quantum-Classical Framework proves that quantum optimisation can be integrated into existing institutional workflows without waiting for future hardware revolutions.

The Result:

- ✓ - Superior risk-adjusted performance (34% volatility reduction)
- ✓ - Strict adherence to practical trading constraints
- ✓ - Fully explainable investment logic